

The many times of action observation on corticospinal excitability: Coupling online facilitation with after-effects.

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Does observation leave a trace?

Online motor resonance is well established [1]. Does it leave a measurable corticospinal trace **after observation**?

We tested whether online corticospinal facilitation during observation of upper-limb actions predicts subsequent resting excitability.

Experimental Design

32 healthy right-handed adults Within-subject VR-TMS design

1 BASELINE REST

Rest in VR

Black screen with fixation cross; same seated posture and headset configuration as in the observation phase.

2 ACTION OBSERVATION (AO)

VR + single-pulse TMS during observation

4 randomized conditions

Ego 2D

Action on virtual screen (egocentric, 2D)

Ego 3D

Immersive egocentric action (3D)

Allo 2D

Action on virtual screen (allocentric, 2D)

Allo 3D

Immersive allocentric action (3D)

29 TMS stimuli per condition

3 POST-AO REST

After-effect assessment

Same VR rest scene as baseline; post-observation corticospinal excitability measured.

TMS

left M1

EMG

ECR / FDI

Measures & Analysis

Online facilitation = AO / pre-AO rest

After-effect = post-AO rest / pre-AO rest

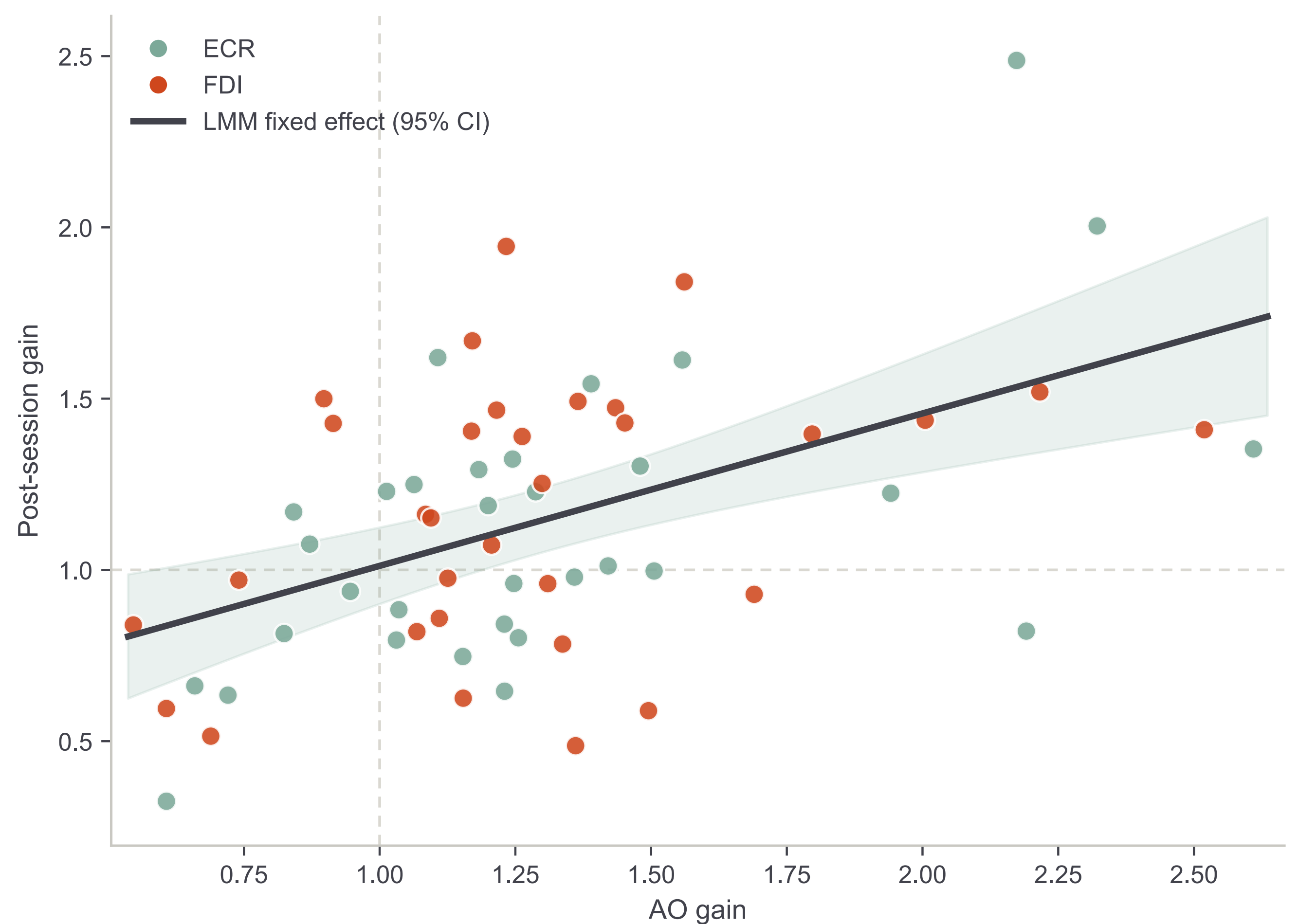
LMMs · random intercept: Subject

Stronger online facilitation predicts stronger post-session corticospinal excitability

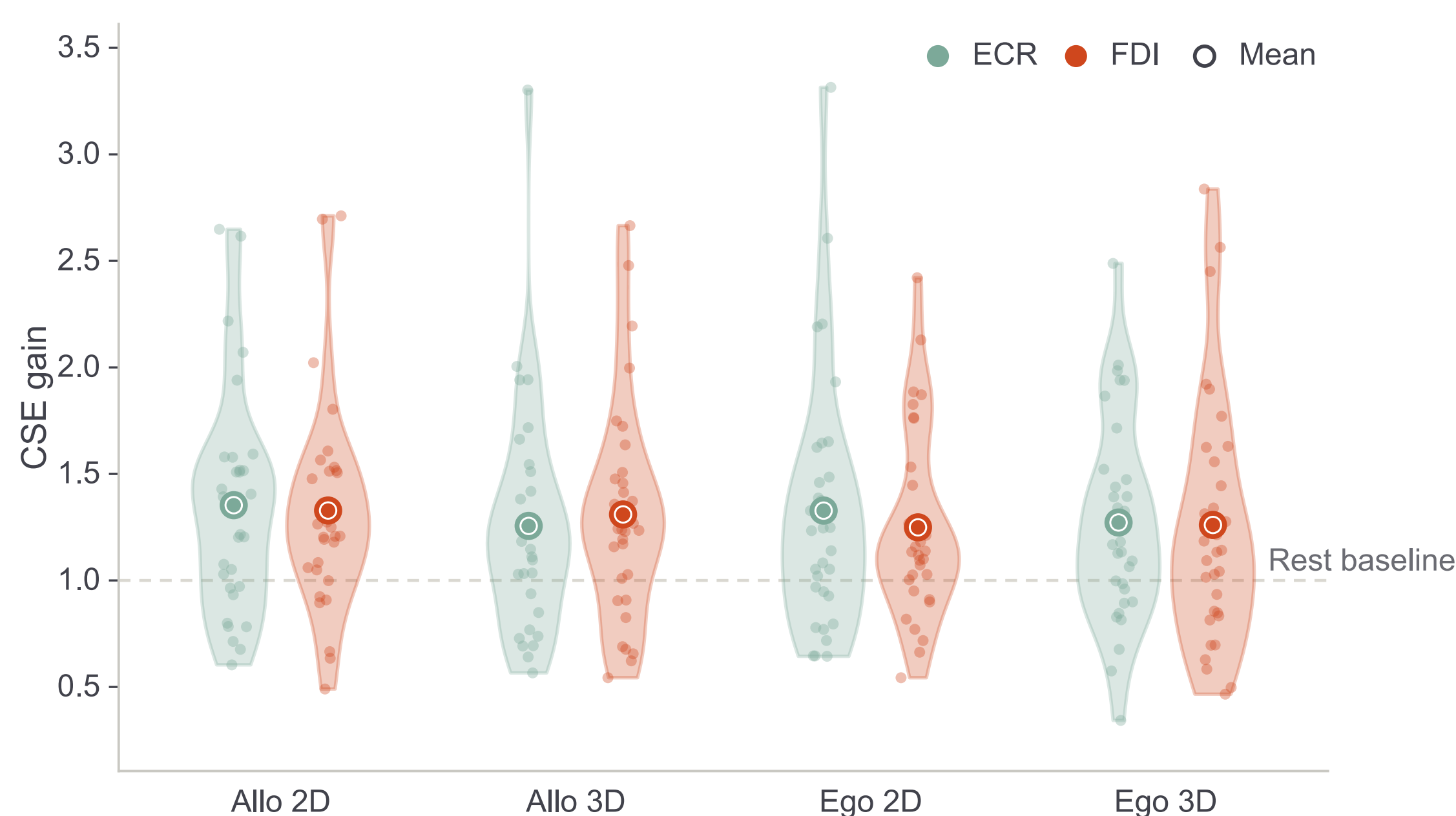
$b = 0.445$

$p < .001$

$R^2_m = .247$



AO reliably facilitates online corticospinal excitability (CSE)



≈30% higher CSE

during AO vs rest

ECR +30.2% · FDI +28.7%
both $p < .001$

No detectable modulation of online CSE by visual factors

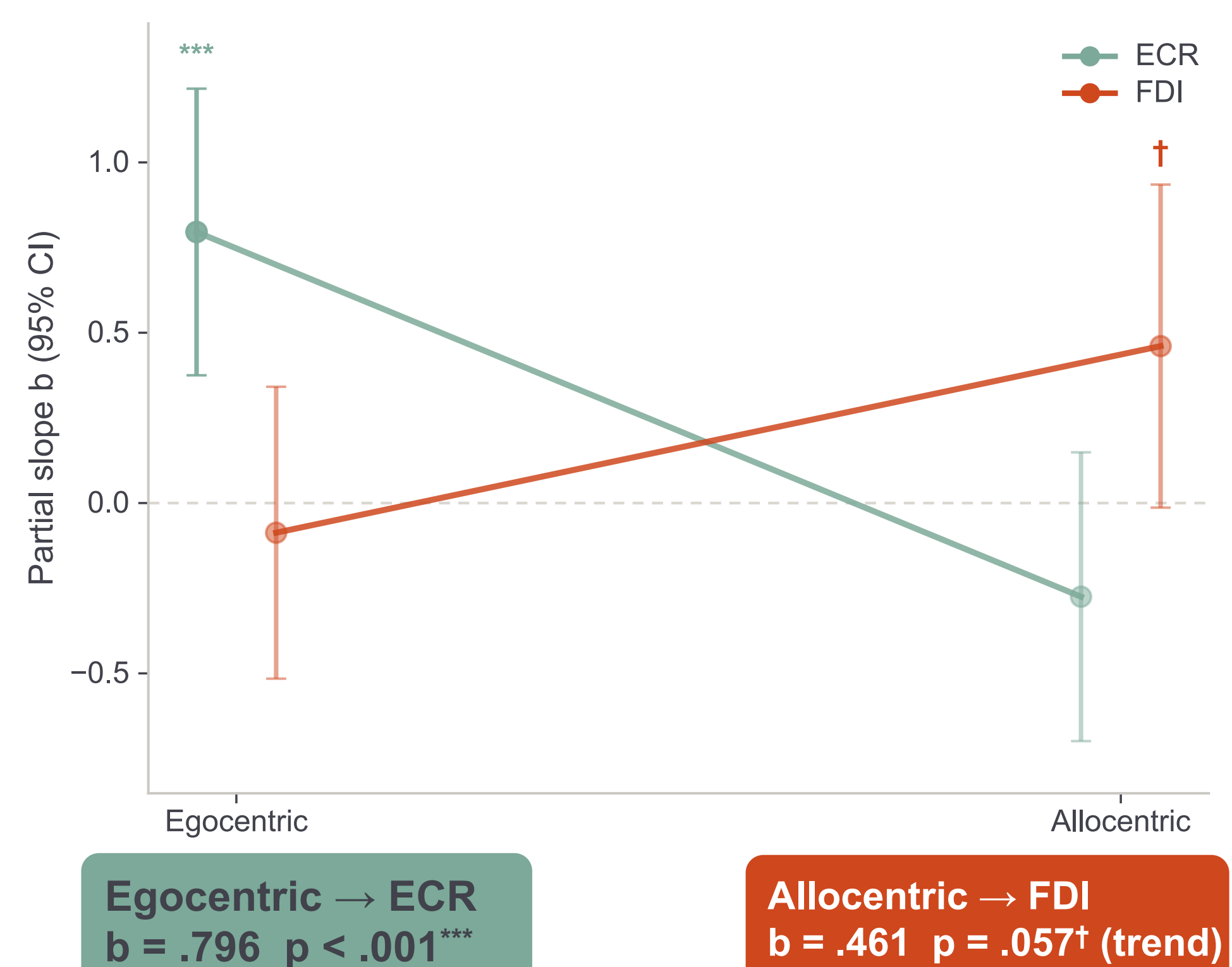
Perspective: *Allo* vs *Ego* · $p = .391$

Dimension: *2D* vs *3D* · $p = .319$

Perspective × Dimension $p = .657$

The online–after relationship shows perspective–muscle specificity

Perspective-dependent muscle modulation
Post-hoc Ego vs Allo contrast: $\Delta b = -1.62$ · $p = .006$



Egocentric → ECR
 $b = .796$ $p < .001^{***}$

Allocentric → FDI
 $b = .461$ $p = .057^{\dagger}$ (trend)

Motor resonance may extend beyond observation.

Stronger online AO facilitation predicts stronger post-session CSE.

This may reflect a carry-over of motor resonance into the post-observation period, although sham-TMS control is needed to disentangle AO-specific from TMS-related contributions.

AOT IMPLICATION

Perspective–muscle specificity may help guide protocol design.

The online–post coupling motivates testing whether the post-observation period can be leveraged to optimize subsequent motor practice.

Reference

1. Naish KR, Houston-Price C, Bremner AJ, Holmes NP. Neuropsychologia. 2014;64:331–348.

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Poster & Abstract

